

Business Requirements Document (BRD)

Document Overview

This Business Requirements Document (BRD) is based on the provided RAW JSON input. The document aims to outline the business requirements, needs, and expectations derived from the given metadata.

Introduction

The input RAW JSON contains a single key-value pair with the key "raw_text" and a corresponding value that represents a normalized text. The absence of detailed metadata or specific technical details in the input suggests that the primary focus is on understanding the requirements related to text normalization or processing.

Business Needs and Objectives

1. **Text Normalization**: The presence of "normalized text" within the input suggests a business need for text normalization. This could be crucial for data consistency, data integration, or text analysis tasks.
2. **Data Processing**: The input implies a requirement for processing or handling text data. This could involve cleaning, transforming, or analyzing text.
3. **Metadata Handling**: Although the input is minimal, the ability to handle metadata from various sources (as indicated by the task description) is a significant business need. This includes being capable of processing metadata from Excel, XML, ETL mappings, datasets, system configurations, and logs.

Functional Requirements

1. **Text Normalization Functionality**: The system should be capable of normalizing text. This could involve converting text to a standard format (e.g., lowercase, removing special characters, handling non-English characters).
2. **Metadata Extraction and Processing**: The system should be able to extract and process metadata from various file types and data sources. This includes Excel files, XML documents, ETL (Extract, Transform, Load) mappings, datasets, system configuration files, and log files.
3. **Flexibility and Scalability**: Given the variety of potential input sources and formats, the system should be flexible and scalable to handle different types and sizes of metadata.
4. **Data Integrity and Consistency**: The system must ensure that the processed data maintains integrity and consistency, especially when dealing with large datasets or complex metadata.

Non-Functional Requirements

1. **Performance**: The system should be able to process metadata efficiently, handling large volumes of data without significant performance degradation.
2. **Security**: The system must ensure the security of the processed data, adhering to relevant data protection standards and regulations.

3. **Usability**: The system should provide an intuitive interface for users to upload/input metadata, configure processing options, and view results.
4. **Compatibility**: The system should be compatible with various data sources and file formats, ensuring broad applicability.

User Stories

1. As a data analyst, I want to be able to upload metadata from different sources (e.g., Excel, XML) and normalize the text data, so that I can ensure data consistency for analysis.
2. As a system administrator, I need the system to handle large volumes of metadata from various sources efficiently, so that we can integrate and process our data effectively.

Acceptance Criteria

1. The system successfully normalizes text data from the input metadata.
2. The system can extract and process metadata from a variety of file types and data sources.
3. The system maintains data integrity and consistency during processing.
4. The system performs well under large data volumes.
5. The system is secure and adheres to relevant data protection regulations.

Assumptions and Dependencies

- The availability of diverse metadata sources for testing.
- Access to relevant data protection standards and regulations for compliance.
- Technical expertise for implementing the text normalization and metadata processing functionalities.

Risks and Mitigation Strategies

1. **Risk**: Incompatibility with certain data sources or file formats.
 - **Mitigation**: Implement flexible data handling mechanisms and continuously update the system to support new formats.
2. **Risk**: Performance issues with large datasets.
 - **Mitigation**: Optimize processing algorithms and consider distributed processing or cloud-based solutions.

Conclusion

The provided RAW JSON input, despite its simplicity, indicates a need for a robust system capable of handling text normalization and metadata processing from various sources. This BRD outlines the key business needs, functional and non-functional requirements, and other critical aspects necessary for developing such a system.